# Evaluation of the learned and optimising components

Generated by `pnpm report` at 2026-10-20T14:33:58.358Z from commit `d8f1e28c5d662596397043aa8d8f7e31601dfb1f`. Every number is recomputed on its fixed seed; the raw values are in `docs/results/`. Charts of the same runs are in `docs/charts/`.

All data is synthetic and says so: the message corpus is hand written, the flights are generated from physical rules with noise. Each component has a held out split chosen by a fixed seed before training, and the agent has no write path to the corpus, the seed or the threshold. None of these components signs, pays or changes a cap.

## 1. Message classifier (`ml/gate.ts`)

Logistic regression on hashed character 1 to 3 grams, 2^14 buckets, trained at module load. It adds a risk score the rule gate can only turn into a hold.

| Metric | Value |
|---|---|
| Training messages | 262 |
| Held out messages | 87 |
| Accuracy | 93.1% |
| Precision | 100.0% |
| Recall | 81.3% |
| F1 | 0.897 |
| False alarm rate | 0.0% |
| Hold threshold | 0.8 |
| Seed | 20251020 |

87 of 349 messages held out by seed 20251020, never trained on; the 52 scripted attack texts are excluded from training. Precision 100.0% (95% CI 87.1 to 100.0, 26 of 26), recall 81.3% (95% CI 64.7 to 91.1, 26 of 32), false alarm 0.0% (95% CI 0.0 to 6.5, 0 of 55). On the 52 unseen scripted attack texts the classifier alone labels 27 as attacks and would hold 13 at threshold 0.8; the 52 of 52 red team result is the full pipeline (rule gate, classifier, policy engine, contract mirrors).

## 2. Telemetry oracle (`ml/plausibility.ts`, `fusion/kalman.ts`, `fusion/conformal.ts`)

Robust z over five features of a flight log, the GPS track smoothed by a Kalman filter with a Rauch Tung Striebel pass, the flag threshold set by split conformal calibration for a 2% false alarm target.

| Metric | Value |
|---|---|
| Flights | 2000 generated, 500 held out, 72 of them tampered |
| Detection | 100.0% |
| False alarm | 3.0% |
| Conformal threshold | z 2.60 for a 2.0% target, 634 calibration flights |

| Metric | Value |
|---|---|
| Coverage | 97.0% |
| Raw track false alarm at matched detection | 13.8% |
| Smoothed track false alarm at matched detection | 0.0% |
| Seed | 7331 |

| Tamper class | Flights | Caught |
|---|---|---|
| inflated_mu | 15 | 15 |
| replayed_track | 16 | 16 |
| impossible_speed | 16 | 16 |
| battery_too_low | 15 | 15 |
| teleport_jump | 10 | 10 |

500 of 2000 synthetic flights held out by seed 7331, 72 tampered; fit on 750, conformal calibration on the honest rows of the next 750. Detection 100.0% (95% CI 94.9 to 100.0, 72 of 72); false alarm 3.0% (95% CI 1.8 to 5.1, 13 of 428) on 428 honest flights of the 500 held out (seed 7331), full five feature model plus replay and teleport checks at the conformal threshold z 2.597. The 0.0% smoothed figure is an ablation on the two track features at a fixed z 4, not the oracle's false alarm rate.

## 3. Lease strategy (`economics/mdp.ts`)

A finite horizon Markov decision process over lease balance, bond, weather and offer class, solved by value iteration, compared with the greedy rule on the same seeded seasons with common random numbers.

| Metric | Greedy rule | MDP policy |
|---|---|---|
| Mean season profit after instalments | 7,471 yuan | 7,540 yuan |
| Standard deviation | 4,882 yuan | 4,837 yuan |
| Seasons reaching title | 84.0% | 84.0% |

46,800 states, 26 weeks, 200 seasons, seed 2026, advantage 69 yuan a season.

## 4. Bond stress test (`economics/stress.ts`)

| Metric | Value |
|---|---|
| Seasons | 2000 |
| Default probability | 9.0% |
| Lender loss given default, mean, no bonds | 21,081 yuan |
| Lender loss given default, p95, no bonds | 25,593 yuan |
| Lender loss given default, mean, with bonds | 14,582 yuan |
| Lender loss given default, p95, with bonds | 19,739 yuan |
| Share of the p95 loss the bonds absorb | 23.0% |
| Expected vendor bond drawdown a season | 358 yuan |

| Metric | Value |
|---|---|
| Expected pilot slashes a season | 346 yuan |
| Seed | 9001 |

## 5. Strategy ratchet (`ratchet.ts`)

| Metric | Value |
|---|---|
| Proposals | 60 |
| Kept | 5 |
| Broke the reserve rule | 10 |
| Profit on the frozen week, baseline | 246 yuan |
| Profit on the frozen week, best | 926 yuan |
| Seeds | eval 4242, walk 99 |

## 6. Pilot credit score (`credit/score.ts`)

Buhlmann credibility on one fixed sequence: the fleet prior, three confirmed jobs, one held log. Score path 0.912, 0.922, 0.930, 0.936, 0.882; final tier B, instalment share 2815 bps inside the band.

## 7. Drift monitor (`drift/cusum.ts`)

One sided CUSUM on the classifier's risk score (target 0.1, k 0.25, h 2). alarm 6 observations after the shift, 0 false alarms in 600 benign scores, synthetic scores on seed 20251020.

## 8. Second price planner (`planner.ts`, `economics/pricing.ts`)

second price: the higher margin per mu flies first and pays the runner up's bid in priority, so an inflated fee cannot buy a slot at a profit.